AMIT MANJARLY

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TECHNICAL SKILLS

ML & Deep Learning: CNNs, ResNet, U-Net, Open-Set Recognition, Causal ML, Conformal Prediction, FNO / PINNs
LLMs & GenAI: Llama-3.2, Mistral, RAG, Fine-Tuning (QLoRA, rsLoRA), Hugging Face, ChromaDB
Computer Vision: YOLOv8/v11, OOD Detection, TorchGeo, Object Detection, Image Segmentation, CV Models
ML Frameworks: PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, XGBoost, LightGBM, NumPy, Pandas
Infra & MLOps: Docker, Kubernetes, MLflow, DVC, CUDA, Apple MPS, FastAPI, Streamlit, Google Earth Engine
Languages & DBs: Python (Advanced), SQL, C++; PostgreSQL, InsForge, SQLite, Hive, Impala, Oracle

EXPERIENCE

Resilient Jul 2026 - Present
AI/ML Software Engineer Intern – Tempe, AZ

- Built physics-informed hybrid surrogates under severe climate stress, cutting prediction RMSE by 86.7% vs a baseline.
- Applied Double ML orthogonalization to cut causal bias by 98.1%, valuing $4.46M in avoided losses across portfolios.
- Downscaled foundation climate models over complex terrains, cutting temperature RMSE by 67.3% to hit 78.0% accuracy.
- Deployed Meta SAM 2 and watershed boundary segmentation for parcel mapping, achieving 0.95 IoU across revisits series.

Accenture PLC Mar 2021 - Jul 2024
Software Engineer – Hyderabad, India

- Scaled enterprise Python test automation frameworks with Tricentis qTest and CI/CD, reducing manual QA cycles by 40%.
- Hardened high-volume Impala SQL database queries across 20M+ loan records, reducing pipeline data delay times by 33%.
- Governed Oracle and Hive batch ETL pipelines through Toad for Oracle, accelerating CCAR loan model throughput by 15%.
- Maintained validation rules and automated quality checks for CECL credit loss models, securing downstream reliability.
- Diagnosed Linux server workloads via SuperPuTTY to eliminate bottlenecks, driving a 25% efficiency gain across loops.
- Architected production Tableau reports for Risk Analytics Platforms to govern pipeline telemetry and portfolio health.
- Institutionalized data engineering standards and led technical workshops, unifying modular development across 4 teams.

PROJECTS

Frontier LLM Inference & Training Platform [GitHub] Oct 2026
PyTorch · Hugging Face (Transformers, PEFT) · CUDA / MPS · Asyncio · FastAPI · Pydantic · Prometheus · NVLink · W4A16 Quantization

- Architected continuous-batching engine with DynamicCache recycling, slashing memory padding waste from 68.4% to 24.1%.
- Engineered SFT pipeline with ChatML prompt masking QLoRA cut peak VRAM by 79.6% (1.85 GB) on Qwen-0.5B at no loss.
- Designed zero-memory DPO reference forward pass via adapter toggling, eliminating duplicate models with +0.310 margin.
- Modeled Ring-AllReduce scaling across NVLink vs PCIe, projecting 89.5% vs 55.9% 8-GPU efficiency and W4A16 trade-offs.

Multi-Agent Coordination & Scaling Platform [GitHub] Oct 2026
Distributed Causal Tracing · MVCC Snapshot Isolation · Anti-Entropy State Sync · Wald's SPRT · Adaptive Traffic Shaping · NetworkX/SciPy

- Formulated swarm scaling laws via extended Amdahl models, discovering a 7.7x peak throughput inflection point at N=64.
- Hardened N=64 swarm engine via vector-clock OCC and dual quorums, resolving 417 write collisions with zero corruption.
- Modeled swarm compute budget allocations, demonstrating a 25/55/20 plan-exec-verify split yields 3.2x cost efficiency.
- Designed SCEP eval to isolate pass@k artifacts, proving +0.58 super-additive synergy on complex asymmetric benchmarks.

Open-World Continual Learning for Autonomous Agents [GitHub] Jan 2026 - Apr 2026
OpenCV · 2D Object Detection · Uncertainty Quantification (UQ) · TFRecords · Protocol Buffers · CUDA Acceleration · MLflow · Docker

- Attained 100% route retention and 5x faster adaptation (31.1s to 6.2s) on novel cities via continual Actor-Critic EWC.
- Trained YOLO11m on 4k frames balanced 1:1:1:1, securing 37.3% mAP@50 and 34.1% recall across 4 held-out AV classes.
- Tuned logit energy threshold to <=5% in-distribution FPR, reaching 95% TPR on held-out unmapped obstacles and hazards.
- Deployed OOD-triggered braking in multi-head policy, preventing collisions across 20 simulated novel hazard scenarios.

EDUCATION

M.S. in Data Science, Analytics and Engineering Aug 2024 - May 2026
Arizona State University – Tempe, AZ
Coursework: Machine Learning, Deep Learning, NLP, Statistical Inference, Artificial Intelligence

B.E. in Computer Engineering Aug 2015 - Jun 2019
Gujarat Technological University – Ahmedabad, India
Coursework: Calculus, C++, Java, System Programming, DSA, OS, DBMS, Computer Networks, OOP